

SCREENING SUMMARY											
Review method: Structured rapid literature review											
Records reviewed during initial title and abstract screening: 60											
Records excluded during initial screening: 7											
Duplicate records removed: 2											
Records retained as potentially relevant: 51											
Records not advanced because full text could not be accessed: 10											
Full-text records assessed: 41											
Sources included in the core evidence set: 18											
Sources retained as reserve literature: 4											
Sources excluded following full-text assessment: 19											
The ten records without accessible full text were not assessed individually because sufficient evidence could not be extracted or verified. Given the rapid and exploratory scope of the review, these records were documented in the initial screening stage but were not advanced to the final evidence set.											
Source ID	Title	Search Area	Full Text Confirmed?	Direct Relevance	Primary Contribution	Evidence Strength	Transferable to Residential Windows?	Overlap with Stronger Source?	Final Decision	Exclusion Reason or Inclusion Justification	Relevant Pages or Sections
A-GS-03	Sustainable Maintenance Strategy for Wood Windows Defects	Window repair and service life	Yes	High	Defect classification and condition-based method for selecting maintenance strategies for existing wooden windows	Moderate	Yes	No	Include	Directly examines wooden-window deterioration and connects observed defect severity with maintenance and component-replacement decisions, although the proposed interventions were not implemented or monitored.	Pages 914-919; defect classifications on pages 916-917; methodology and workflow on pages 918-922; case analysis on pages 925-933
A-GS-05	Serviceability Fragility Functions for New Zealand Residential Windows	Window repair and service life	Yes	Low	Experimental evidence on residential-window gaps, jamming, frame rotation and repair needs following seismic wall movement	Moderate	Partly	No	Exclude	The study provides credible evidence on window damage and repair requirements, but the failure mechanism is earthquake-specific and does not directly inform ordinary maintenance, component replacement or product-life extension.	Pages 137-142; experimental method; damage criteria; fragility functions; conclusions and research limitations
A-GS-06	The Re-Seal Window: A Redesign of the Edge Seal of Insulated Glass Units to Facilitate Easy and Fast Remanufacturing	Design for disassembly	Yes	High	Component-level redesign that allows the short-life edge seal of an insulated glazing unit to be replaced while retaining the glass and spacer components	Moderate	Yes	No	Include	Directly investigates how a failure-prone window component can be separated and replaced to support remanufacturing and product-life extension, although the design remains exploratory and some sealing tests were inconclusive.	Pages 9-11; pages 23-29; pages 32-49; preliminary design and testing on pages 64-102; final design and implementation on pages 103-147; conclusion and recommendations on pages 157-164
A-GS-08	Steel Window Restoration: Effective Strategies for Restoring and Upgrading Historic Steel Windows	Window repair and service life	Yes	High	Practice-based guidance for assessing, disassembling, repairing, restoring, thermally upgrading and, where necessary, replacing historic steel windows	Moderate	Partly	No	Include	Provides detailed practical evidence on the conditions under which steel-window restoration remains technically and economically realistic, including repair methods, component reuse, thermal upgrades and circumstances in which complete replacement may be justified.	Pages 2-4: restoration-versus-replacement assessment and condition survey; pages 4-7: removal sequence, corrosion repair, coatings, hardware and glazing; pages 7-9: weatherstripping, IGU/VIG retrofit and storm-window upgrades; page 10: conclusion

A-GS-11	Life Cycle Perspectives of Fixed and Operable Wooden Windows	Lifecycle comparison	Yes	High	Fifty-year lifecycle comparison of fixed and operable wooden windows, including glazing, hardware, coatings, aluminium cladding and maintenance	High	Yes	Partial	Include	Provides transparent lifecycle evidence showing how window complexity, glazing, aluminium cladding and maintenance choices affect environmental impacts, while also identifying burden shifting between different impact categories.	Pages 3-7; lifecycle scope and assumptions on pages 4-7; results on pages 7-10; discussion and limitations on pages 11-14; conclusions on pages 14-15
A-GS-17	Optimizing the Performance of Window Frames: A Comprehensive Review of Materials in China	Window repair and service life	Yes	Low	Broad review of wood, polymer, aluminium and composite frame materials, degradation mechanisms and performance-improvement strategies	Moderate	Partly	Yes	Exclude	The paper provides useful general background on material degradation and maintenance, but it does not directly analyse repair, component replacement or lifecycle extension and substantially overlaps with more focused included sources.	Sections 2-3; material degradation; improvement methods; maintenance and service-life discussion; conclusion
A-GS-18	Life Cycle Assessment of a Wood-Aluminium Window According to European Norms	Lifecycle comparison	Yes	Moderate	Standards-based whole-life environmental baseline for a European wood-aluminium window, including production, transport, operational heat loss, end of life and recycling credits	High	Yes	Partial	Exclude	The study provides a rigorous lifecycle baseline and identifies important environmental hotspots, but it excludes maintenance, repair, refurbishment and individual component replacement, so its final value must be compared with the more directly relevant Search D evidence.	Pages 1-3; goal, scope and exclusions on pages 2-4; inventory and scenario assumptions on pages 5-8; results and interpretation; conclusions on pages 13-15
B-GS-01	Interconnections: An Analysis of Disassemblable Building Connection Systems towards a Circular Economy	Design for disassembly	Yes	Moderate	Built case-study evidence on accessible, concealed and reversible structural and non-structural connections in a modular building designed for complete disassembly and relocation	High	Partly	Yes	Exclude	The study provides credible practical evidence and achieved complete disassembly with very little waste, but its modular-building connection evidence substantially overlaps with B-GS-04. B-GS-04 is retained because it more directly explains the additional ownership, information, inspection, repair and value-retention conditions required to turn disassembly into actual reuse.	Pages 2-6: research questions, methodology and case boundaries; pages 6-15: connection systems and disassembly solutions; pages 15-17: outcomes, conclusions and limitations
B-GS-02	Design for Circular Disassembly of High-Performance Facade Systems	Design for disassembly	Yes	High	Window-specific disassembly framework combining component mapping, connection analysis, disassembly sequencing, time estimation, recovery assessment and digital product data	Moderate	Yes	No	Include	Directly analyses fixed and operable aluminium window systems and identifies how permanent sealants, rolled thermal breaks, inaccessible components and hardware affect disassembly, reuse and material recovery. It also provides quantitative disassembly results and practical criteria that can inform the preliminary screening framework.	Pages 1-4: research context and comparison of disassembly methods; pages 5-6: window case and component-network analysis; pages 6-7: data structure and application workflow; pages 7-8: quantitative results, limitations and conclusion
B-GS-03	Coding the Circularity: Design for the Disassembly and Reuse of Building Components	Design for disassembly	Yes	Moderate	Conceptual translation of disassembly principles into parametric design rules covering functional separation, assembly sequence, lifecycle coordination, connection accessibility, joint geometry and component standardisation	Limited	Yes	Yes	Exclude	The paper provides a useful conceptual list of disassembly criteria, but it does not test or validate the proposed parametric rules through a window or building-component case. Its main criteria are already covered more rigorously by the validated B-GS-19 method and the façade-specific B-GS-20 framework.	Pages 1-3: research aim and literature basis; pages 3-7: macro-criteria, relational models and parametric design objectives; page 8: conclusions

B-GS-04	Disassembly and Reuse of Demountable Modular Building Systems	Design for disassembly	Yes	Moderate	Empirical evidence on the complete disassembly-to-reuse process, including connection release, disassembly sequencing, take-back arrangements, digital tracking, quality inspection, repair and component replacement	High	Partly	Partial	Reserve - not included in core analysis	Provides strong real-world evidence showing that technically demountable connections do not automatically make disassembly easy and that successful reuse also depends on ownership, information access, traceability, inspection and value-retention services. These system conditions are transferable to the repair and reuse of window components, although the case concerns a modular building.	Pages 1-6: research context and aims; pages 16-20: mixed-method case-study design and data analysis; pages 21-38: disassembly sequence, take-back mechanism, tracking, inspection and repairs; pages 39-41: conclusions and limitations
B-GS-05	Robotic Disassembly and Window Upgrading	Design for disassembly	Yes	High	Window-component mapping and disassembly-sequence methodology for replacing glazing while retaining aluminium-frame components	Moderate	Yes	Partial	Include	Directly investigates an aluminium window and demonstrates how a disassembly map can organise the removal, repair, replacement and upgrading of individual components. The evidence remains exploratory because the physical demonstration used a scaled model and the robotic gripping process required further development, but the disassembly logic is highly relevant.	Pages 1-3: window upgrading, repair and disassembly-map principles; pages 4-5: case-study method and virtual simulation; pages 5-7: component sequence, robotic demonstration, results and conclusion
B-GS-06	Design-for-Disassembly Principles for Apartment Blocks - Experimental Design in a Finnish Housing Project	Design for disassembly	Yes	Moderate	Research-by-design comparison of conventional and DfD apartment-building solutions using reuse rate, component batch size, standardisation and structural independence	Moderate	Partly	Yes	Exclude	The study provides useful quantified evidence on standardisation, simplicity and independent building systems, but it focuses mainly on concrete load-bearing structures and assumes that reversible joints will become technically feasible. It does not test window systems and overlaps with the more thoroughly validated B-GS-19 disassembly framework.	Pages 1-4: research aim and design principles; pages 4-9: case modification and indicators; pages 9-11: quantified results, limitations and further research
B-GS-07	Advancing Circular Economy Strategies in Modular Architecture through Design for Disassembly and Reuse	Design for disassembly	Yes	Low	Secondary synthesis of modular-construction strategies, digital tools, circular business models and international DfD case studies	Limited	Partly	Yes	Exclude	The paper provides a broad summary of modular design for disassembly, but it relies mainly on publicly available secondary information and does not provide sufficiently transparent or independently verified evidence. Its design principles, implementation barriers and environmental claims are already covered more rigorously by the included empirical and lifecycle studies.	Pages 1-5: literature discussion and modular DfD strategies; pages 6-9: case-study comparison, methodological limitations and implementation barriers; pages 9-10: conclusions and future research
B-GS-09	Increasing Reuse Potential by Taking a Whole Life-Cycle Perspective on the Dimensional Coordination of Building Products	Design for disassembly	Yes	High	Façade-specific method linking window and curtain-wall dimensions, functional requirements, production efficiency and compatibility with future reuse contexts	Moderate	Yes	No	Reserve - not included in core analysis	Adds a distinct condition that is not sufficiently covered by the other disassembly studies: a component may be physically removable but still difficult to reuse when its dimensions, glazing characteristics or configuration are incompatible with a new building. The façade and window case therefore provides directly transferable evidence for assessing future re-applicability.	Proceedings PDF pages 27-29: research problem, reuse-potential model and method; pages 29-33: façade dimensions, functional configurations, glass-production yield and case-study results; page 34: discussion, limitations and conclusion

B-GS-10	Systems for Reuse, Repurposing and Upcycling of Existing Building Components	Design for disassembly	Yes	Moderate	Systems-level component-management model combining early component audits, information exchange, demand identification, storage, testing, recertification and triage between reuse, repurposing, upcycling and lower-value recovery	High	Partly	No	Include	Provides strong evidence that technical recoverability alone is insufficient because component reuse also depends on timely information, visible demand, storage, logistics, quality assurance, certification, procurement and supportive policy. These conditions are directly relevant to whether repairable or reusable window components can realistically remain in circulation, even though the study is not specific to windows.	Pages 43-64: barriers, reuse markets and management options; pages 66-87: mixed-method research design; pages 89-103: case-study evidence and component-management model; pages 106-143: information system, triage framework and live case studies; pages 184-193: conclusions, recommendations and limitations
B-GS-11	Taxonomy Supporting Design Strategies for Reuse of Building Parts in Timber-Based Construction	Design for disassembly	Yes	Moderate	Literature-based taxonomy connecting design for disassembly and adaptability with building layers, building-part scales and reversible connectors	Moderate	Partly	Yes	Exclude	The taxonomy provides useful terminology, but it is limited to timber-based design, has not been validated through practical cases and does not provide direct technical evidence on window repair or component replacement. Its conceptual contribution also substantially overlaps with B-GS-12 and with stronger included disassembly frameworks.	Pages 1-5: aim, scope and literature method; pages 6-10: building layers, parts and reuse terminology; pages 10-14: proposed taxonomy and practical interpretation; pages 14-15: conclusions and limitations
B-GS-12	Reversible Architecture: Reuse of Timber Building Parts in Circular Design	Design for disassembly	Yes	Moderate	Broad synthesis of reversible timber design supported by stakeholder interviews, literature reviews, barriers analysis and a taxonomy of design-for-reuse strategies	Moderate	Partly	Yes	Exclude	The thesis contains useful evidence on information needs, reversible connectors, standards, markets, certification and stakeholder barriers, but it is broad, timber-specific and substantially overlaps with B-GS-10 and the article recorded as B-GS-11. The stronger and more focused sources provide the required evidence without retaining this entire thesis.	Pages 12-16: abstract, aims and scope; pages 22-37: reuse, building layers, connectors and DfD principles; pages 37-45: methods and stakeholder interviews; pages 46-55: barriers, drivers and taxonomy; pages 56-61: discussion and conclusions; pages 81-90 and 93-108: included papers
B-GS-13	Building Design for Sustainability: Towards Zero-Waste Buildings-Building Design for Reuse and Disassembly	Design for disassembly	Yes	Low	Architectural exploration combining general DfD principles, reference projects, material mapping and a catalogue of design solutions for reusable building components	Limited	Partly	Yes	Exclude	The thesis contains useful architectural illustrations and design ideas, including proposals for windows and doors, but the proposed solutions were not technically tested or validated. Its contribution is mainly exploratory and overlaps with stronger sources that provide tested assessment methods and empirical evidence.	Pages 4-10: aim, scope and method; pages 18-30: reuse barriers and DfD principles; pages 40-50: building and material mapping case study; pages 53-72: design catalogue, including envelope, windows and doors; pages 74-77: demonstration structure and discussion
B-GS-14	Towards a Sustainable Circular Economy: Understanding the Environmental Credits and Loads of Reusing Modular Building Components from a Multi-Use Cycle Perspective	Lifecycle comparison	Yes	High	Multi-use-cycle lifecycle assessment of component reuse, repair, replacement, recycling and disposal, supported by uncertainty and global sensitivity analysis	High	Partly	Partial	Include	Directly examines whether component reuse continues to provide environmental benefits once repair, replacement, transport and future-use uncertainties are included. It is especially relevant because it distinguishes direct reuse, repairable components and replaceable components, showing that high replacement requirements can substantially reduce the expected benefit of reuse.	Pages 1-3: research problem, aims and value-retention processes; pages 3-8: allocation methods, component model and lifecycle assumptions; pages 8-13: environmental results, uncertainty and sensitivity analysis; pages 13-16: interpretation, design implications and conclusions

B-GS-15	Integrated Platform-Based Tool to Improve Life Cycle Management and Circularity of Building Envelope Components	Design for disassembly	Yes	Moderate	Prototype digital platform integrating building-envelope data, lifecycle assessment, defect records, supplier information, material tracking and dismantling guidance	Moderate	Partly	Yes	Exclude	The source is relevant to building-envelope information management, but the platform remains at an early validation stage and focuses mainly on demonstrating digital functions. Its useful conditions concerning documentation, tracking and dismantling information are already covered by B-GS-10 and B-GS-20, which are more directly connected to the final screening framework.	Pages 1-7: research problem, platform components and façade demonstration cases; pages 7-18: user requirements, lifecycle data and digital functions; pages 18-24: prototype testing and validation; pages 24-27: discussion, limitations and conclusions
B-GS-16	Design for Disassembly as an Alternative Sustainable Construction Approach to Life-Cycle-Design of Concrete Buildings	Design for disassembly	Yes	Moderate	Detailed evaluation and redesign of precast concrete systems and façades using functional separation, lifecycle coordination, assembly sequence, connection accessibility and component-level disassembly criteria	Moderate	Partly	Yes	Exclude	The doctoral thesis provides a comprehensive and technically detailed DfD assessment, but it focuses on large precast concrete structural and façade systems rather than residential windows or replaceable window components. Its assessment criteria are already represented by the more transferable B-GS-19 method and the façade-specific B-GS-20 framework.	Pages 9-27: DfD theory and assessment model; pages 75-109: evaluation of precast structural, infill and façade systems; pages 111-141: development and application of demountable façade systems; pages 143-145: conclusions and recommendations
B-GS-17	An ICT-Enabled Product Service System for Reuse of Building Components	Design for disassembly	Yes	Moderate	Proof-of-concept integration of RFID, BIM, cloud-based component tracking and product-service business models, demonstrated through an internal glazed partition system	Moderate	Yes	Yes	Exclude	The paper provides a relevant example of how supplier ownership, maintenance services, performance data and digital tracking could support the extended use of glazed components. However, it remains a short proof-of-concept study and its main system conditions are covered more comprehensively by B-GS-10, while its digital-platform contribution overlaps with B-GS-15.	Pages 1-2: Product-Service System concept and research method; pages 2-5: BIM, RFID, data platform and glazed-system demonstration; pages 5-6: potential benefits, conclusions and future development
B-GS-18	Life Cycle Assessment of a Swedish Multifamily Building Designed for Disassembly and Flexibility: Impact of Allocation Methods on Future Scenarios	Lifecycle comparison	Yes	Moderate	Whole-building lifecycle comparison of linear and circular future scenarios using 100:0, 50:50 and system-expansion allocation methods	High	Partly	Yes	Exclude	The study provides rigorous evidence that allocation choices and future scenarios can substantially affect circular lifecycle results. However, it evaluates whole-building transformation rather than repair or replacement of individual components, while B-GS-14 and the window-specific Search D studies are more directly aligned with the planned lifecycle scenario comparison.	Pages 1-4: circular lifecycle problem and allocation approaches; pages 4-12: case study and scenario method; pages 13-18: lifecycle and sensitivity results; pages 18-21: implications, limitations and conclusions
B-GS-19	Metrics for Building Component Disassembly Potential: A Practical Framework	Design for disassembly	Yes	Moderate	Validated and streamlined quantitative method for assessing component disassembly potential using connection accessibility, deconstructability, material quantities and lifecycle environmental data	High	Yes	No	Include	Provides one of the strongest methodological foundations for the screening framework because the proposed method was tested through expert assessment, statistical comparison and three building case studies. It is not window-specific, but its component-level criteria and integration with BIM and lifecycle data are directly transferable to residential window assemblies.	Pages 2-4: review of existing methods and research gap; pages 4-10: research design, expert Delphi process and proposed assessment method; pages 10-17: statistical validation, case-study results and limitations; page 17: conclusions

B-GS-20	Disassembly Assessment Framework for Enhanced Reclamation of Façade Systems	Design for disassembly	Yes	High	Façade-specific framework combining disassembly potential, material circularity, reclamation potential, disassembly time and structured component databases	Moderate	Yes	Partial	Include	Provides a detailed façade-specific method and demonstrates how connection choices affect component reclamation, reuse potential and disassembly time. The comparison between clamped, silicone-bonded and gasket-based solutions is especially useful for window systems, although parts of the assessment depend on subjective ratings, assumptions and estimated disassembly times.	Pages 24-45: circular façade lifecycle, barriers and enabling conditions; pages 47-61: review of assessment indicators and disassembly mapping; pages 63-89: framework and assessment modules; pages 92-110: curtain-wall case study, scenario comparison and proposed improvement; pages 112-118: conclusions, limitations and recommendations
C-GS-01	Is Prolonging the Lifetime of Passive Durable Products a Low-Hanging Fruit of a Circular Economy? A Multiple Case Study	Lifecycle comparison	Yes	High	Five real product cases combining lifecycle assessment and manufacturer lifecycle costing to test whether repair and refurbishment of passive durable products reduce environmental impacts and costs	High	Yes	Partial	Include	Provides directly transferable evidence for passive durable components such as residential windows. The five cases show that life extension can reduce both environmental impacts and manufacturer costs when production dominates the lifecycle, while also demonstrating that the result depends on the number of additional uses, the environmental burden of the retained component, service activities, transport and the limit beyond which repeated repair may no longer remain beneficial.	Pages 1-4: research context and conditions for environmentally beneficial lifetime extension; pages 4-7: case-study method, LCA and LCC boundaries; pages 7-15: product cases, lifecycle results and sensitivity analysis; pages 16-21: environmental and economic interpretation, rebound risks, limitations and conclusions
C-GS-02	Circular Economy Rebound	Rebound effects and unintended consequences	Yes	High	Foundational definition of circular-economy rebound based on the extent to which secondary production substitutes for primary production, including substitution and price mechanisms	Moderate	Yes	Partial	Include	Provides the essential theoretical basis for identifying a genuine circular-economy rebound effect. It explains that repair, reuse, refurbishment or recycling only achieve their expected environmental benefit when secondary production displaces primary production and does not increase total production or consumption through limited substitution, lower prices or expanded demand.	Pages 1-3: circular-economy rationale, displacement problem and definition of secondary production; pages 3-5: relationship with energy rebound and market-based definition of circular rebound; pages 5-7: substitution, price and production mechanisms; pages 7-8: proposed mitigation approaches, implications and conclusions
C-GS-03	The Circularity Paradox: The Rebound Effect, Governance, and the Limits of Corporate Sustainability	Rebound effects and unintended consequences	Yes	High	Updated systematic review of 75 studies linking circular strategies with direct, indirect, psychological and systemic rebound effects, governance failures, insufficient indicators and mitigation through sufficiency	High	Yes	Partial	Reserve - not included in core analysis	Provides an updated and broad synthesis of rebound risks across reuse, repair, remanufacturing, recycling and access-based models. It adds a distinct governance and sufficiency dimension by showing that technological efficiency and circular business practices may legitimise additional consumption when they are not supported by appropriate indicators, production limits, cultural change, regulation and multilevel governance.	Pages 1-6: research problem, rebound concepts and relationship with circular strategies; pages 6-10: systematic-review method and bibliographic portfolio; pages 10-23: rebound types, conditioning factors, governance failures and mitigation instruments; pages 23-27: integrative framework, practical implications, limitations and conclusions

C-GS-04	The Rebound Effect of Circular Economy: Definitions, Mechanisms and a Research Agenda	Rebound effects and unintended consequences	Yes	High	Systematic-review framework distinguishing initiating, developing and mitigating mechanisms of circular-economy rebound across individual, organisational and policy levels	High	Yes	Partial	Include	Provides the most directly usable mechanism framework for the qualitative codebook and preliminary screening tool. It distinguishes the initial circular intervention from the behavioural, market, infrastructure and production mechanisms that develop the rebound and from the measures intended to mitigate it, while also emphasising contextual dependence, system boundaries and the need for early detection.	Pages 1-3: circular-economy and rebound definitions, boundaries and research questions; pages 3-4: systematic-review protocol and qualitative coding method; pages 4-8: initiating, developing and mitigating mechanisms; pages 8-11: major lessons, early-detection needs, research agenda, limitations and conclusions
C-GS-06	Evaluating Lifetime Extension Strategies to Close the Loop: Comparing the Material Circularity and Environmental Life Cycle Performance of Traction Batteries	Lifecycle comparison	Yes	Moderate	Combined LCA and Product Circularity Indicator comparison of repair, component replacement, remanufacturing and repurposing scenarios for traction batteries	Moderate	Partly	Yes	Exclude	The thesis provides a detailed component-level scenario method and useful evidence that the benefits of life extension depend on additional lifetime, component condition and the burden of replacement parts. However, the results are strongly shaped by battery chemistry, state-of-health thresholds, declining use-phase energy efficiency and battery-specific recycling assumptions. Its transferable insights are already covered by B-GS-14, C-GS-01 and C-GS-10, which are more suitable for the residential-window study.	Pages 13-29: repair, component replacement, remanufacturing and implementation challenges; pages 29-39: LCA, circularity method, assumptions and scenarios; pages 40-47: lifecycle, circularity and sensitivity results; pages 48-54: strategy hierarchy, trade-offs, limitations and conclusions
C-GS-07	Impacts, Synergies, and Rebound Effects Arising in Combinations of Product-Service Systems and Circularity Strategies	Rebound effects and unintended consequences	Yes	Moderate	Ex-ante screening of seven combinations of product-service systems and circular strategies using business, environmental and social impact and rebound-effect frameworks	Limited	Partly	Yes	Exclude	The study demonstrates that combinations of repair, maintenance, upgrading, leasing and other strategies may create both synergies and additional rebound risks. However, it is a short exploratory conference paper based on expert assessment within one business-to-business capital-goods company, and the predicted impacts were not measured after implementation. Its useful concepts substantially overlap with the stronger C-GS-04 mechanism framework.	Pages 1-2: research gap and single-case methodology; pages 2-4: configuration selection and business, environmental and social screening; pages 4-5: synergies, costs and potential rebound effects; page 6: conclusions, limitations and further research
C-GS-08	Careless Product Use in Access-Based Services: A Rebound Effect and How to Address It	Rebound effects and unintended consequences	Yes	Moderate	Empirical evidence that reduced product care can shorten product lifetimes in access-based services, supported by a user survey, design-expert input and interviews with service providers	Moderate	Partly	No	Reserve - not included in core analysis	Adds a distinct behavioural mechanism that is not covered adequately by the lifecycle and market-based sources. The study demonstrates that expected lifetime-extension benefits can be reduced when users care less for products and shows how information, reminders, monitoring, maintenance support, incentives and business-model design may improve product care. Transferability to windows is partial because the evidence concerns leased washing machines, but the findings remain relevant to maintenance behaviour and user acceptance.	Pages 1-3: access-based services, product care and behavioural-rebound hypothesis; pages 3-4: user survey and comparison of accessed and owned products; pages 4-8: design-expert study, provider interviews and ten product-care strategies; pages 8-9: business-model implications, limitations and future research

C-GS-09	A Study of the Rebound Effect on the Product-Service System: Why Should It Be a Top Priority?	Rebound effects and unintended consequences	Yes	Moderate	Conceptual frameworks for locating potential product-service-system rebound effects across lifecycle stages, sustainability dimensions and responsible actors	Limited	Partly	Yes	Exclude	The paper provides a useful lifecycle structure for asking when and where rebound effects may occur, but the framework is illustrated through a simulated car-sharing example and has not been validated through measured outcomes. Its classification of rebound causes and lifecycle stages overlaps with the more rigorous and directly applicable C-GS-04 framework.	Pages 1-2: systematic-review method and product-service-system rebound concepts; pages 3-4: lifecycle and deployment frameworks; pages 4-5: simulated car-sharing application; pages 5-6: conclusions, limitations and validation needs
C-GS-10	Towards Sustainable Development Through the Circular Economy-A Review and Critical Assessment on Current Circularity Metrics	Rebound effects and unintended consequences	Yes	High	Critical review of circularity indices and assessment frameworks showing how narrow material-flow metrics can conceal environmental, economic or social burden shifting	High	Yes	Partial	Include	Provides a strong methodological basis for separating circularity from environmental performance. It demonstrates that a high circularity score does not by itself prove sustainability and supports combining material or component indicators with lifecycle assessment, transparent system boundaries and explicit allocation rules. These principles are directly relevant to the 50-year scenario comparison and to preventing burden shifting from being incorrectly classified as a successful circular outcome.	Pages 1-3: review aims, method and validity requirements; pages 3-8: classification and explanation of circularity indices and assessment tools; pages 9-11: comparison against environmental, economic and social requirements; pages 11-13: burden-shifting risks, allocation issues, recommendations, limitations and conclusions
D-GS-01	The Influence of Durability and Recycling on Life Cycle Impacts of Window Frame Assemblies	Lifecycle comparison	Yes	High	Comparative 80-year lifecycle assessment of aluminium, wood, aluminium-clad wood and PVC window frames across alternative maintenance, component-replacement, service-life and recycling-allocation scenarios	High	Yes	Partial	Include	Directly demonstrates that assumptions about maintenance, hardware and seal replacement, frame service life and end-of-life recycling can substantially change the calculated environmental performance of window systems. It provides a strong methodological basis for the scenario comparison, although its maintenance cycles are modelled assumptions, glazing and operational energy are excluded, and its material rankings should not be treated as universally applicable.	Pages 1-4: research problem, goals, functional unit and system boundaries; pages 4-7: window assemblies, component data, maintenance regimes and recycling scenarios; pages 7-11: maintenance, service-life and end-of-life results; pages 11-13: interpretation, limitations and conclusions
D-GS-02	Maintaining or Replacing a Building's Windows: A Comparative Life Cycle Study	Lifecycle comparison	Yes	High	Sixty-year case-study comparison of maintaining existing windows, replacing and maintaining new wooden windows, and replacing with wooden, aluminium-clad or PVC windows using lifecycle climate assessment, lifecycle costing and building-energy simulation	High	Yes	Partial	Include	Provides highly direct evidence that maintaining existing residential windows can produce lower lifecycle climate impacts and costs than replacement, while showing that the result changes under specific conditions involving heating-system emissions, initial window performance, service life and intervention frequency. It is retained alongside D-GS-03 because it gives more detailed evidence on alternative replacement-window materials and a single real housing case.	Pages 1-4: research context, previous evidence and research questions; pages 4-8: case building, intervention options, LCA, LCC and energy-simulation method; pages 8-16: lifecycle results and sensitivity analyses; pages 16-20: discussion, conclusions, implications and limitations

D-GS-03	A Comparative Life Cycle Study of Window Interventions: Impact of Building Characteristics and Local Context	Lifecycle comparison	Yes	High	Comparative lifecycle climate and cost assessment of maintenance, replacement, maintained replacement and secondary-glazing options for two contrasting residential buildings, supported by sensitivity analyses of building, location and methodological parameters	High	Yes	Partial	Include	Adds essential evidence that there is no universally preferable window intervention because results depend on the building's initial thermal performance, size and age, as well as heating emissions, climate, transport availability, intervention frequency and reference period. Although it extends the earlier D-GS-02 study and therefore partly overlaps with it, it contributes a distinct cross-building and context-sensitive decision framework that is directly relevant to identifying conditions and rebound-related trade-offs.	Pages 1-5: research gap, aims and lifecycle boundaries; pages 5-10: two building cases, intervention options, LCA, LCC, energy simulation and sensitivity design; pages 10-17: comparative results and sensitivity analyses; pages 17-21: discussion, context-dependent findings, conclusions and limitations
D-GS-08	Comparative Whole Building Life Cycle Assessment of Renovation and New Construction	Lifecycle comparison	Yes	Low	Whole-building methodology for comparing renovation with new construction by separating existing, reused and newly added components and defining which lifecycle modules should be included	High	Partly	Yes	Exclude	The paper provides a rigorous and useful explanation of lifecycle boundaries, including the need to account for maintenance, repair and replacement of retained components. However, its case concerns whole-building adaptive reuse rather than window repair or component replacement, and its methodological contribution is already sufficiently represented by the more directly relevant component and lifecycle sources retained from Searches B, C and D.	Pages 1-5: renovation-LCA terminology, research gap and lifecycle stages; pages 5-10: comparison of alternative system boundaries and proposed assessment scope; pages 10-18: adaptive-reuse case study, inventory method and results; pages 18-20: methodological limitations and conclusions
D-GS-09	Embodied Life Cycle Assessment Comparison of Residential Building Retrofit Measures in Atlanta	Lifecycle comparison	Yes	Low	Embodied-impact comparison of multiple retrofit measures for four residential-building archetypes, including window replacement, insulation, shading and HVAC upgrades	Moderate	Partly	Yes	Exclude	The study confirms that window replacement can create substantial embodied impacts and that manufacturing and transportation burdens should not be ignored. However, windows are only one of many retrofit measures, the study does not compare window maintenance, repair or component replacement, recurring maintenance is excluded and operational impacts are addressed only through approximate payback calculations. The more window-specific studies provide stronger and more complete evidence for this project.	Pages 1-8: research aim, literature review and identified evidence gap; pages 8-16: residential archetypes, retrofit options and embodied-LCA method; pages 16-23: embodied-impact and energy-payback results; pages 24-27: policy implications, limitations and conclusions

D-GS-10	Life Cycle Operating Energy Saving from Windows Retrofitting in Heritage Buildings Accounting for Technical Performance Decay	Lifecycle comparison	Yes	High	One-hundred-year comparison of window maintenance, draught proofing, secondary glazing, glazing replacement and complete window replacement while explicitly modelling the deterioration and replacement frequency of seals, coatings and insulated glazing units	High	Partly	Partial	Include	Provides unique evidence that the expected performance of a window intervention is not constant throughout its service life and that component durability, performance decay and maintenance frequency can alter the long-term benefit. The combination of onsite blower-door testing, infrared thermography, calibrated simulations and 120 scenarios is highly valuable for the screening framework, although the study concerns a heritage building and assesses operational energy rather than a complete environmental lifecycle including embodied impacts.	Pages 1-4: research problem, intervention scenarios and technical-performance decay; pages 4-9: onsite measurements, window modelling and deterioration assumptions; pages 9-17: maintenance, draught-proofing, secondary-glazing, glazing-replacement and complete-replacement scenarios; pages 17-19: comparative results, limitations and conclusions
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